OPEN DATA "P:\School\Master\Econometrics of Finance\Projet\2025\Test.xlsx"

CALENDAR(D) 1990:1:1

DATA(FORMAT=XLSX,ORG=COLUMNS) 1990:01:01 2021:07:23 BEL<<"BEL-DS" US<<"US-DS" BTC<<"USD-BIT"

*suppression des jours à partir du 23/07/2021

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* modèle des rendements boursiers

set P1 = BEL

set P2 = US

set P3 = BTC

set R1 = log(P1) - log(P1{1})

set R2 = log(P2) - log(P2{1})

set R3 = log(P3) - log(P3{1})

*Méthode GARCH avec Constante

GARCH(P=1,Q=1,ITERS=500) / R1 R2

Constant

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MV-GARCH - Estimation by BFGS
Convergence in    53 Iterations. Final criterion was  0.0000081 <=  0.0000100

Daily(5) Data From 1990:01:02 To 2021:07:23
Usable Observations                     8234
Log Likelihood                   54240.4887

    Variable                    Coeff      Std Error      T-Stat      Signif
*****************************************************************************
1.  Mean(R1)                 6.27154e-04  1.06204e-04      5.90516  0.00000000
2.  Mean(R2)                 5.66268e-04  9.10857e-05      6.21687  0.00000000

3.  C(1,1)                   1.90387e-06  2.36299e-07      8.05703  0.00000000
4.  C(2,1)                   4.99430e-07  8.71229e-08      5.73248  0.00000001
5.  C(2,2)                   1.84726e-06  2.77594e-07      6.65454  0.00000000
6.  A(1,1)                       0.04816      0.00393     12.24923  0.00000000
7.  A(2,1)                       0.03535      0.00368      9.59828  0.00000000
8.  A(2,2)                       0.07479      0.00724     10.32664  0.00000000
9.  B(1,1)                       0.93629      0.00520    180.01166  0.00000000
10. B(2,1)                       0.94541      0.00520    181.88773  0.00000000
11. B(2,2)                       0.90548      0.00917     98.75391  0.00000000
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*Méthode VAR-GARCH

GARCH(P=1,Q=1,MV=VECH,REGRESSORS,ITERS=1000) / R1 R2

Constant R1{1} R2{1}

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MV-GARCH, VECH - Estimation by BFGS
Convergence in   366 Iterations. Final criterion was  0.0000000 <=  0.0000100

Daily(5) Data From 1990:01:03 To 2021:07:23
Usable Observations                     8233
Log Likelihood                   54847.0079

    Variable                    Coeff      Std Error      T-Stat      Signif
*****************************************************************************
Mean Model(R1)
1.  Constant                0.000497045  0.000089888      5.52959  0.00000003
2.  R1{1}                  -0.030755190  0.012859792     -2.39158  0.01677614
3.  R2{1}                   0.272419572  0.012344021     22.06895  0.00000000
Mean Model(R2)
4.  Constant                0.000702742  0.000084925      8.27487  0.00000000
5.  R1{1}                   0.021415611  0.011450471      1.87028  0.06144467
6.  R2{1}                  -0.032230747  0.012118026     -2.65974  0.00782020

7.  C(1,1)                  0.000025267  0.000001182     21.38425  0.00000000
8.  C(2,1)                  0.000012765  0.000000711     17.95529  0.00000000
9.  C(2,2)                  0.000019735  0.000000670     29.43978  0.00000000
10. A(1,1)(1)               0.034728229  0.006730700      5.15968  0.00000025
11. A(1,1)(2)               0.043461539  0.011437607      3.79988  0.00014477
12. A(1,1)(3)               0.078494566  0.009394387      8.35548  0.00000000
13. A(2,1)(1)               0.017898692  0.003491138      5.12689  0.00000029
14. A(2,1)(2)               0.028937383  0.006106969      4.73842  0.00000215
15. A(2,1)(3)               0.037730814  0.004860516      7.76272  0.00000000
16. A(2,2)(1)               0.033083414  0.005288788      6.25539  0.00000000
17. A(2,2)(2)              -0.006357752  0.009976954     -0.63724  0.52396605
18. A(2,2)(3)               0.092593697  0.007490789     12.36101  0.00000000
19. B(1,1)(1)              -0.069861126  0.021523663     -3.24578  0.00117128
20. B(1,1)(2)               1.501564058  0.058201344     25.79947  0.00000000
21. B(1,1)(3)               0.059844886  0.048257187      1.24012  0.21492961
22. B(2,1)(1)              -0.542528699  0.023966715    -22.63676  0.00000000
23. B(2,1)(2)               1.750308755  0.046866017     37.34708  0.00000000
24. B(2,1)(3)               0.034600941  0.025733194      1.34460  0.17875334
25. B(2,2)(1)              -0.815455871  0.041497909    -19.65053  0.00000000
26. B(2,2)(2)               1.193508651  0.051166060     23.32618  0.00000000
27. B(2,2)(3)               0.996477504  0.041974392     23.74013  0.00000000
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